Practical No. 04

Title: Collaboration in Git

- Create a New Branch using git checkout -b branchname
- Design more figma screens and put it in local repo folder
- Make changes to a file (e.g., README.md):
- Add and commit the changes
- Push the Branch to Remote
- Create a Pull Request (PR) on github
- Merge the PR

Procedure

Step 1: Create a Repository on GitHub

1. Open www.github.com and sign in.
2. Click **New repository**.
3. Enter the repository name: `Mobile-App-UI`.
4. Select Public or Private visibility.
5. Check **Add a README file**.
6. Click **Create repository**.

A new repository named `Mobile-App-UI` is created on GitHub.

Step 2: Clone the Repository

1. Open the GitHub repository.
2. Click the green **Code** button.
3. Select HTTPS and copy the repository URL.
4. Open Git Bash.
5. Navigate to the practical folder:

```
cd /f/GitPractical/Git-Pr-04
```

6. Clone the repository by using the following command:

```
git clone https://github.com/USERNAME/Mobile-App-UI.git
```

Replace `USERNAME` with your GitHub username.

7. Enter the project directory:

```
cd Mobile-App-UI
```

8. Open the project in VS Code:

```
code .
```

9. Check the repository status:

```
git status
```

The GitHub repository is copied to the local computer.

Step 3: Create a New Branch

A branch is a separate line of development used to make changes without directly modifying the main branch.

1. Open the terminal in VS Code.
2. Check the existing branches:

```
git branch
```

3. Create a new branch and switch to it:

```
git checkout -b figma-screens
```

4. Verify the current branch:

```
git branch
```

A new branch named `figma-screens` is created and activated.

Step 4: Add Images to the Local Repository

1. Open the project in VS Code.
2. Create a new folder named `figma-screens`.
3. Copy two or three sample images from the computer.
4. Paste the images into the `figma-screens` folder.

Example folder structure:

```
Mobile-App-UI/
├── README.md
└── figma-screens/
    ├── login.png
    ├── home.png
    └── profile.png
```

Sample images are added to the local project repository.

Step 5: Modify the README.md File

The `README.md` file contains information about the project.

1. Open `README.md` in VS Code.
2. Add or update the following content:

```
# Mobile App UI Design

## Project Description
This project contains sample mobile application UI screens.

## Screens
- Login Screen
- Home Screen
- Profile Screen

## Tools Used
- Git
- GitHub
- Visual Studio Code
```

3. Save the file by pressing `Ctrl + S`.

The project documentation is updated.

Step 6: Add Changes to the Staging Area

The `git add` command prepares modified and new files for committing.

1. Check the changes:

```
git status
```

2. Add all changes to the staging area:

```
git add .
```

3. Verify the staged changes:

```
git status
```

The modified README file and new images are staged for committing.

Step 7: Commit the Changes

The `git commit` command saves the staged changes in the local Git history.

Execute the following command:

```
git commit -m "Add sample images and update README"
```

Verify the repository status:

```
git status
```

If all changes have been committed, Git may display:

```
nothing to commit, working tree clean
```

The changes are saved in the local repository.

Step 8: Push the Branch to GitHub

The `git push` command uploads the local branch and its committed changes to the remote repository.

Execute:

```
git push -u origin figma-screens
```

The `figma-screens` branch is uploaded to GitHub.

Step 9: Create a Pull Request (PR)

A Pull Request is a request to review and merge changes from one branch into another.

1. Open the `Mobile-App-UI` repository on GitHub.
2. Click the **Pull requests** tab.
3. Click **New pull request**.
4. Select the following branches:
    - Base: `main`
    - Compare: `figma-screens`
5. Enter the title:

    ```
    Add sample images and update README
    ```

6. Write a short description explaining the changes.
7. Click **Create pull request**.

A Pull Request is created to merge the changes into the main branch.

Step 10: Merge the Pull Request

1. Open the created Pull Request.
2. Review the changes under **Files changed**.
3. Return to the Conversation tab.

4. Click **Merge pull request**.
5. Click **Confirm merge**.

After a successful merge, the changes from `figma-screens` become part of the `main` branch.